TrueSkate v1 - every net, in plain English

A net is one named electrical connection: all its pads and copper are joined. This guide covers all 67 nets; n means axis 1-6, p means channel 1-3. The carrier is wiring, sockets, capacitors, resistors and a fuse holder; the plugged-in modules do the control.

NET / FAMILY	CONNECTION AND PURPOSE
12V_IN	J5 positive to F1 input. Raw motor-supply power, before the fuse. Wrong polarity can damage this board.
12V_FUSED	F1 output to all six Mini VIN pads and both capacitors per axis. The fuse interrupts sustained excess input current; it does not regulate motor current.
3V3	Teensy 3.3 V output to encoder contact 5. USB powers the Teensy; neither the carrier VIN contact nor any Mini regulator output feeds this net.
GND	Zero-volt reference for Teensy, encoders and bridge GND pads. Joins supply negative and all motor returns at copper junction NT1. USB and motor supply are not electrically isolated.
PGND1 ... PGND6	One separate motor-power return per axis: Mini GND power pad, its capacitor negatives and EN pull-down. The six returns meet GND only at NT1 on the bare carrier.
CGND1 ... CGND6	Mini control contacts 1 and 4 to that axis's optional bridge pad. Leave JP1-JP6 OPEN. The installed Mini already joins these contacts internally to its own power return; closing a bridge adds a parallel path through logic ground.
SPI_SCK	Teensy pin 13 to encoder contact 1 on all six sockets. Clock pulses tell encoders when to shift each data bit.
SPI_MOSI	Teensy pin 11 to encoder contact 3. Commands go out from the Teensy to all encoders.
SPI_MISO	Encoder contact 2 to Teensy pin 12. Only the selected encoder may send angle data back.
A<n>_PWM<p> (18 nets)	Three low-power pulse-width-modulation instructions per axis, Teensy to Mini control contacts 3/5/7. All axes use edge-aligned PWM at one common frequency (intended 25 kHz). These are NOT motor-phase power.
A<n>_EN (6 nets)	Teensy to Mini control contact 9. High enables that driver. R21-R26 each connect EN through 10k ohms to its own PGND, holding it low when the Teensy is not driving. This cannot override a stuck-high output and is not a hardware kill.
A<n>_CS (6 nets)	Teensy to encoder contact 4. Chip select is active low: choose one encoder at a time. Firmware must establish all selects high before SPI transfers.
A<n>_PH<p> (18 nets)	Three switched motor-power outputs per axis: Mini OUT1/2/3 (socket P3/P2/P1) to male phase-header contacts 1/2/3. Never move these cables with 12 V live.

TEENSY PIN NUMBERS (not socket contact numbers)

AXIS / MOVEMENT	PWM1 / PWM2 / PWM3	EN	CSn
A1 / index curl 1	4 / 5 / 6	16	26
A2 / index curl 2	2 / 22 / 23	17	27
A3 / index splay	7 / 36 / 29	20	32
A4 / middle curl 1	24 / 0 / 25	21	38
A5 / middle curl 2	3 / 33 / 37	30	39
A6 / middle splay	8 / 9 / 28	31	40

No-connect contacts are NOT nets: Teensy VIN and unused digital contacts; Mini 2 (its regulator output), 6 (reset), 8 (sleep), 10 (fault). Encoder square pad/marked pin 1 is CLK; order is CLK, MISO, MOSI, CSn, 3V3, GND. V1 has no polarity protection, hardware kill or guaranteed capacitor discharge. Disconnect 12 V and measure residual voltage before moving wires. Human review before ordering; this is not a manufacturing release.